

Nicole Ye's Portfolio

[Linkedin](#) | [Nicole Ye's Resume](#) | [My Website!](#) | n7ye@uwaterloo.ca | [Github](#)

If you have any inquiries, please don't hesitate to reach out! :)

SKILLS

- Firmware & Embedded: ESP32, Raspberry Pi Pico, C++, Java, PlatformIO, I2C, UART, SPI
- Electronics: KiCad, EasyEDA, PCB Schematic & Layout, Soldering, Multimeter, Oscilloscope, Power Supply, Cable Management, Crimping
- Sensors & Protocols: VL53L4CX ToF, RDM630 RFID, Hall-Effect, Encoders,
- Mechanical/CAD: Onshape, SolidWorks, BambuStudio, 3-D Printing, Sheet Metal, Snap-Fits

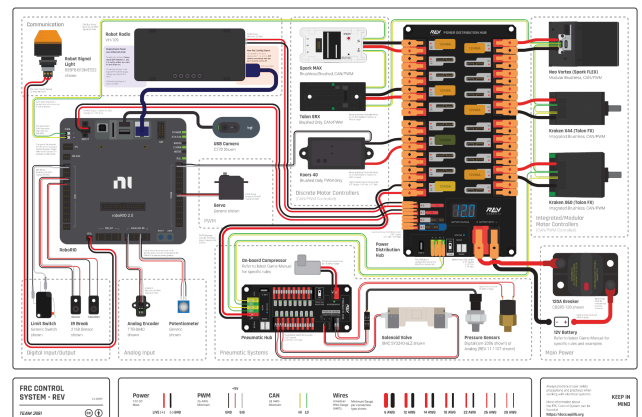
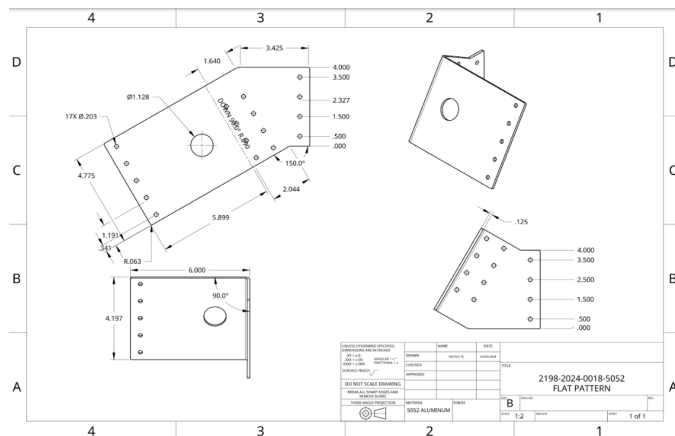
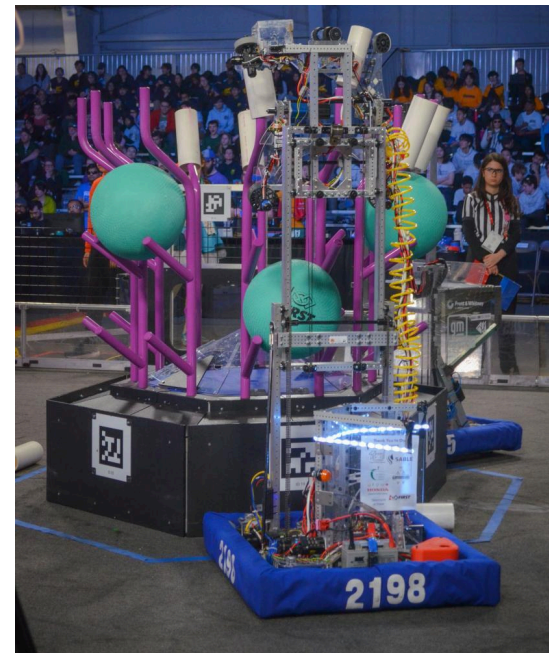
Team 2198 Paradigm Shift Robotics Electrical System

Electrical System Bring-Up

- Utilized [TKO Circuit Simulator](#) to design electrical layout
- Implemented a customized electrical system, including wiring, power distribution, and sensor integration, ensuring compliance with competition standards
- Integrated a line sensor to detect whether a game piece was secured in the holding mechanism, improving automation
- Wired encoders for precise motor control and accurate movement tracking across drive and manipulator systems
- Implemented color sensors to differentiate game piece types and enable autonomous decision-making

Motion Tracking

- Programmed AprilTag-based field-relative localization in Java using Limelight vision co-processor, enabling automated robot alignment to scoring positions and reducing driver repositioning time during matches



Robot Part Drawings OnShape/Solidworks

- Utilized Solidworks and Onshape to create DXF, STEP, and PDF drawings of sheet metal of First Robotics Robots (2021-2025) for CNC processing including standard views, section, and detail dimensions.
- Designed and fabricated gussets for a FIRST Robotics robot, ensuring precise alignment and secure connections between major mechanisms to enhance structural integrity

Prometheus Industries (GreaseTech)

CAD Enclosure Design & DFM

- Engineered a form-fitting enclosure in SolidWorks, precisely modeling around the Milwaukee grease gun chassis for zero interference with grip, trigger, or battery swap.
- Iterated through multiple prototype revisions on a Bambu X1, using rapid print turnaround to stress-test fit, tolerances, and assembly before finalizing the design.
- Built a vibration- and grease-resistant housing with an angled, flush-mounted display bezel — protecting internal PCBs while keeping the UI durable and easy to read in a shop environment.

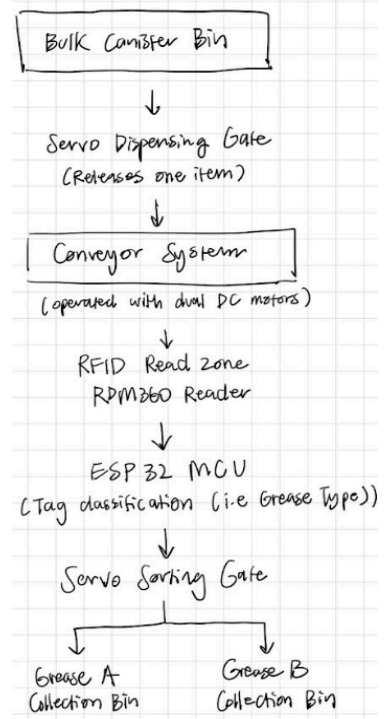
PCB Bring Up

- Designed a custom PCB from schematic through routed layout, integrating dual RFID interfaces, a Hall sensor input, and a relay-based motor driver stage alongside ESP32 UART/OTG connectivity.
- Validated board integrity with multimeter continuity and power-supply testing, confirming clean connections and correct power delivery before assembly.

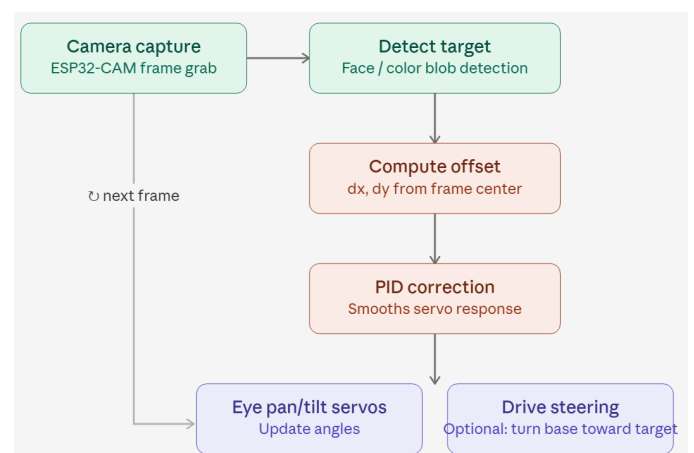
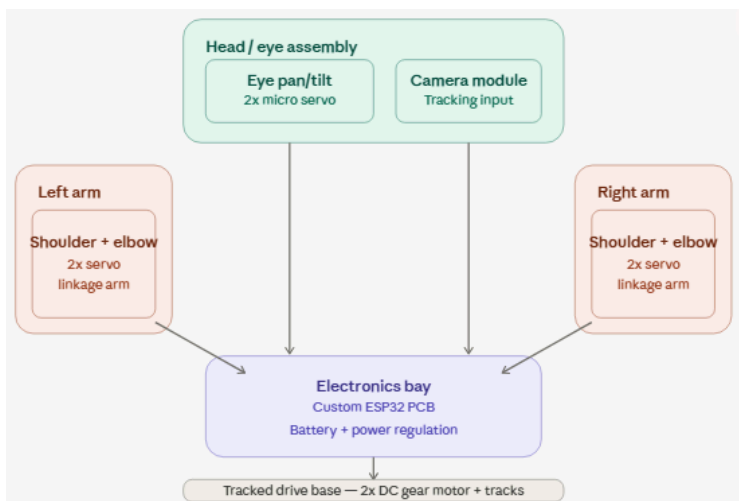


Automated RFID Conveyor Sorting System

- Built an end-to-end automated sorting system that dispenses, identifies, and sorts RFID-tagged canisters with zero operator intervention.
- Integrated an RDM630 RFID reader with an ESP32 to decode tags in real time and classify canisters by grease type via prefix matching.
- Engineered a dual-motor conveyor and servo-driven diverter gate to transport and route canisters into the correct collection bin based on classification.
- Designed embedded control logic to synchronize dispensing, sensing, and sorting into a single deterministic processing cycle.
- Achieved 720 tags/hour sustained throughput with a 5-second dispense-read-sort cycle and consistent classification accuracy.



WALL-E Animatronic Robot (In Progress)



- Designed a custom 2-layer ESP32-based PCB in KiCad integrating USB programming, 3.3V/5V power regulation, a DRV8833 motor driver, connectors for 6 SG90 servos, dual N20 gear motors, and an ESP32-CAM vision module.
- Applied mixed-signal PCB design practices including RF antenna keep-outs, power/logic separation, decoupling networks, bulk capacitance for transient loads, and DRC/manufacturability verification prior to fabrication.
- Performed board bring-up and hardware validation by verifying power rails, flashing firmware through an onboard USB-to-serial interface, and debugging servo and motor outputs using oscilloscopes and multimeters.
- Developed ESP32 firmware enabling Wi-Fi teleoperation through a browser-based interface, implementing real-time control of six servos and differential-drive motors using LEDC PWM and WebSocket/HTTP communication.
- Designed and assembled a fully articulated 3D-printed WALL-E platform featuring tracked locomotion, dual-arm actuation, pan-tilt eyes, integrated battery power, and onboard electronics packaging.
- Implemented machine-vision tracking using an ESP32-CAM to perform color-blob detection, transmitting target offsets over UDP to the main controller for closed-loop servo tracking using proportional/PID control.
- Integrated manual and autonomous operating modes with status indication, battery monitoring, and system documentation including block diagrams, pin mappings, and communication interface specifications.